

A 10b 320MS/s Self-Calibrated Pipeline ADC

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Abstract—A high-speed low-power self-calibrated pipeline ADC is presented. Gain error due to low-gain opamp used in multiplying DAC (MDAC) is corrected by the proposed foreground calibration technique. It adjusts the inter-stage gain by connecting a calibration capacitor into the MDAC positive feedback path. It only requires 168 clock cycles to complete the calibration without external precise references. The calibration circuit does not consume power during normal conversion. The prototype ADC in 90nm low-power CMOS technology achieves conversion rate of 320MS/s with peak SFDR and SNDR of 66.7dB and 54.2dB. The total power dissipation is 42mW and it occupies an active chip area of 0.21mm². Its figure-of-merit (FOM) is 442fJ/conversion-step.

Index Terms—Pipeline analog-to-digital converter (ADC), inter-stage gain, low-gain opamp, self-calibrated.

I. INTRODUCTION

Multimedia and wireless systems are always desired to have higher bandwidth for more information transfer. These systems demand 10b ADCs that allow over hundreds of mega-samples per second (MS/s) with low power and low cost. Opamps, which designed with short channel devices in finer technology, have higher bandwidth, low power, small area, but low dc gain. Insufficient opamp dc gain introduces gain error and nonlinearity to the closed-loop system. Closed-loop MDACs in a 10b pipeline ADC require precise inter-stage gain for residue amplification.

Various calibration techniques were developed to reduce capacitor mismatch effect for higher resolution. Some calibration techniques can also be adapted to calibrate inter-stage gain due to low-gain opamp. Recent foreground calibration work [1] uses least mean square, LMS, engine to adjust inter-stage gain coefficients and high order effect terms due to low-gain opamp, capacitor mismatch, and opamp nonlinearity for each MDAC. However, LMS requires a very long start-up time of the converter to complete calibration. It reports tens of thousands clocks to converge and the off-chip calibration circuit occupies approximately 20,000 gates and consumes 8mW of power. The work [2] applies foreground calibration to estimate the inter-stage gain coefficients first. Then, background calibration during normal conversion is assisted by the estimated gain coefficients. It can greatly reduce self-calibrated clock cycles compared to background calibration alone. Both works require additional MDAC stages because the weighted radix for each output is scaled by the less-than-2 coefficient. Extra power and circuit are

needed for the MDACs and the calibration circuit.

In this paper, a foreground calibration technique is introduced. It corrects gain error due to low-gain opamp by controlling feedback factor. No power is consumed by the proposed calibration circuit during normal conversion. Moreover, the self-calibrated process can be completed within 168 clock cycles, which also greatly reduces the start up time of conversion. In Section II, the self-calibrated algorithm is first introduced. Then, the proposed ADC block diagram and circuit implementation are presented in Section III. The measurement result of the prototype ADC is given in Section IV. Finally, Section V concludes this work.

II. SELF-CALIBRATED ALGORITHM

Negative feedback with high-gain opamp allows accurate operation by suppressing open-loop nonlinearity characteristic. A simple closed-loop gain can be calculated as follows.

$$A_{CL} = \frac{A}{1 + A\beta} = \frac{1}{\beta} \cdot \left(\frac{1}{1 + 1/(A\beta)} \right), \quad (1)$$

where β is the feedback factor, and A is the opamp dc gain. For an ideal opamp with infinite A , A_{CL} is equal to $1/\beta$. However, A_{CL} is smaller than $1/\beta$ with finite A . If β can be adjusted to a little bit smaller, A_{CL} can be calibrated back to the desired value. In a pipeline ADC, residue voltages are amplified precisely by closed-loop MDACs. The gain error due to low-gain opamp in the MDAC can be compensated by adjusting feedback factor.

A. MDAC with Calibration Capacitor

Fig. 1 shows an MDAC with a calibration capacitor, C_{cal} , which capacitance can be digitally adjusted. In the sample phase, eight sample capacitors, C_s , track input. Both C_{cal} and feedback capacitor, C_f , are discharged by connecting to a common node voltage. In the amplify phase, the digital output of sub-ADC connects C_s either to $+V_{ref}$ or $-V_{ref}$. With error overlapping technique, C_f is set to $2C_s$; therefore, it ideally gives an inter-stage gain of 4. C_f is connected as a negative feedback path for the opamp, and C_{cal} is connected as a positive feedback path. It results in the charge subtraction of C_{cal} to C_f . The MDAC output can be written as

$$V_{out} = A_v \left(V_{in} - \sum_{i=1}^8 D_i \frac{V_{ref}}{8} + \frac{8C_s + C_f + C_{cal} + C_p}{8C_s} V_{os} \right), D_i = \pm 1 \quad (2)$$

where C_p is the parasitic capacitor, and V_{os} is the opamp offset voltage. A_V is the MDAC inter-stage gain, and can be expressed as

$$A_V = \frac{8C_s}{C_f + \frac{8C_s + C_f + C_{cal} + C_p}{A} - C_{cal}} \quad (3)$$

In this work, A_V is desired to be 4. The denominator of A_V has a negative term of C_{cal} because C_{cal} is in the positive feedback path. A value of C_{cal} can be found to cancel the $(8C_s + C_f + C_{cal} + C_p)/A$ term. Thus, the purpose of self-calibrated process is to obtain C_{cal} capacitance for A_V approaching 4.

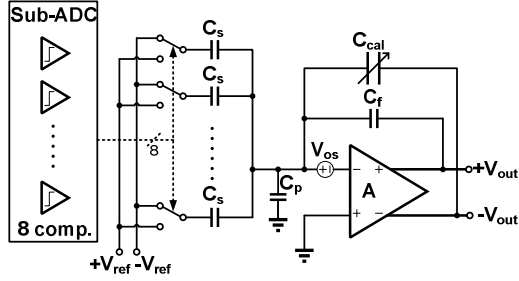


Fig. 1. MDAC with a calibration capacitor.

B. Self-Calibrated Process

Fig. 2 presents a self-calibrated functional diagram for the pipeline ADC. The opamp in the last MDAC is assumed to have enough dc gain to meet accuracy requirement of the last stage sub-ADC; therefore, no calibration procedure is needed. It can be assumed as an “ideal” MDAC. The self-calibrated process starts from the 2nd last MDAC. A $1/8 V_{ref}$ voltage is injected to the under-calibration MDAC at the sample phase. During the amplify phase, the closed-loop MDAC amplifies the $1/8 V_{ref}$ by A_V . The “ideal” MDAC samples the under-calibration MDAC output. Next, the “ideal” MDAC subtracts $3/8 V_{ref}$ and amplifies by a gain of 4. If the under-calibration MDAC inter-stage gain is equal to 4, the “ideal” MDAC output should be $1/2 V_{ref}$. The “ideal” MDAC output is compared with $1/2 V_{ref}$ by a calibration comparator. Then, the comparator output is stored in successive approximation (SA) registers, which digitally control the capacitance of C_{cal} . After a number of cycles, the calibration data stored in the registers would make the gain of the under-calibration MDAC closest to 4. The under-calibration MDAC acts as “ideal” MDAC afterwards. The self-calibrated process moves on to the MDAC stage before it. The self-calibrated process is finished after the 1st MDAC is calibrated. To calibrate 3 MDAC stages, the self-calibrated process needs 56 clock cycles to complete. However, some redundant calibration clock cycles are used to simplify the digital circuit design. Thus, it requires 168 clock cycles to complete the self-calibrated process in this work.

In (2), the $(8C_s + C_f + C_{cal} + C_p)V_{os}/8C_s$ term caused by the opamp offset does not affect the performance during the normal conversion. It is because the offset can be corrected by error overlapping technique. However, during the self-calibrated process, the $1/8 V_{ref}$ input at the under-calibration MDAC is processed through subsequent MDACs and

examined whether the output is $1/2 V_{ref}$. The “ideal” MDACs perform as analog signal processors that amplify and subtract a fixed value. The opamp offset in each MDAC stage would cause a voltage drift. Like the input, it is processed through subsequent MDAC stages and could cause error decision for the self-calibrated process. Input offset storage (IOS) technique [3] is utilized during the calibration to reduce the offset error. It can suppress the input-referred offset voltage, up to 10.5LSB, in the 1st MDAC with the minimum opamp dc gain of 30dB to meet 10b accuracy.

Analog self-calibrated technique usually requires precise reference sources. C_s in the 1st MDAC is designed to have a plane area which has mismatch less than 0.1%. The $1/8 V_{ref}$ input voltage to be injected can be created by MDAC switched-capacitor arrangement as shown in Fig. 3. For the under-calibration MDAC, $4C_s$, $3C_s$, and $1C_s$ sample $+V_{ref}$, $-V_{ref}$ and V_{agnd} respectively in the sample phase. Therefore, the net input sampled voltage is equal to $1/8 V_{ref}$. To implement the IOS technique, the opamp offset is sampled by unity-gain closed-loop configuration. During the amplify phase, two sets of $4C_s$ are connected to $+V_{ref}$ and $-V_{ref}$ respectively for zero subtraction. C_f and C_{cal} are connected to the positive and negative terminals of the opamp respectively to amplify the input sampled voltage. For “ideal” MDAC, all $8C_s$ sample the prior MDAC output in the sample phase. Its opamp offset is also sampled. Then, in the amplify phase, $5C_s$, $2C_s$ and $1C_s$ are connected to $+V_{ref}$, $-V_{ref}$ and V_{agnd} respectively for subtracting $3/8 V_{ref}$. The residue voltage is amplified by closed-loop configuration.

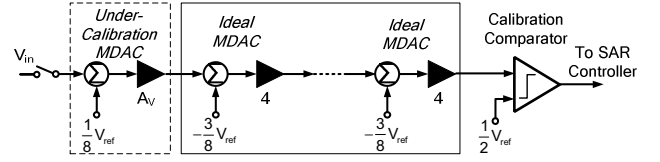


Fig. 2. Self-calibrated functional diagram.

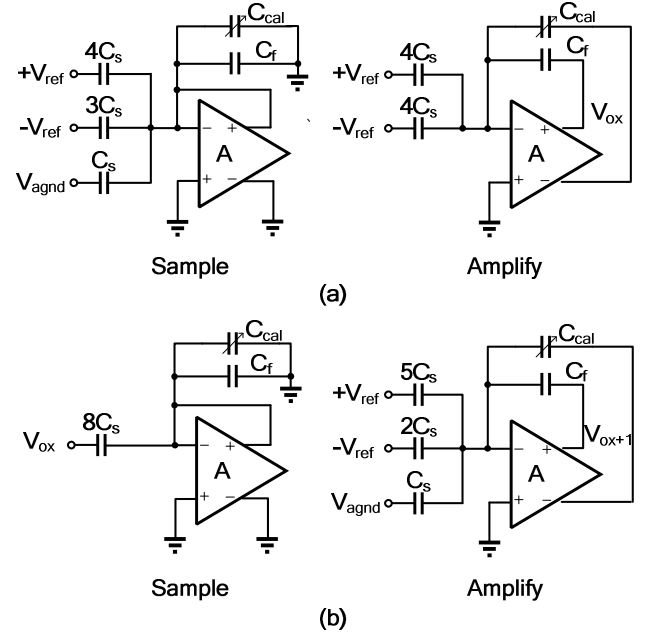


Fig. 3. Switched-capacitor configuration during the self-calibrated process for (a) under-calibration MDAC and (b) ideal MDAC.

III. CIRCUIT IMPLEMENTATION

A. ADC Block Diagram

The system block diagram of this ADC is given in Fig. 4. For low-power operation, the sample-and-hold circuit is merged with the sample switches of the 1st MDAC and sub-ADC. A clock-boost circuit is implemented in the input sample switches of the 1st MDAC to improve the linearity with high speed sampling. The ADC contains 4 stages of 2.5b MDAC and a 2b sub-ADC. Since the opamp in the last MDAC stage gives enough precision for the last stage sub-ADC, only the first three MDACs include calibration capacitor to correct inter-stage gain. Once the self-calibrated process begins, the calibration sequence controller activates MDAC and SA registers. After the self-calibrated process is completed, the calibration data are stored in the 7b SA registers to be used for the normal conversion. An additional calibration comparator is only used in the self-calibrated process, not during the normal conversion.

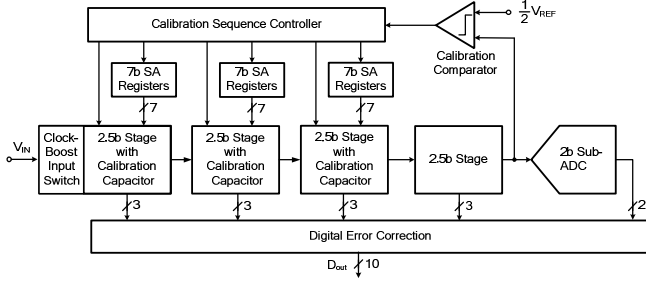


Fig. 4. Proposed ADC block diagram.

B. Calibration Capacitor

Based on ADC resolution requirement, A_V in Eq. (3) can have finite error tolerance. To meet accuracy requirement for 2.5b per stage 10b pipeline ADC, the inter-stage gain error of the 1st MDAC must be less than 0.39%. The schematic of the calibration capacitor C_{cal} is shown in Fig. 5. It is a binary-weighted capacitor array that realizes 128 calibration steps with 7 bit registers. In this design, C_{cal} has one step resolution equal to 0.285% of C_s . The step resolution for the inter-stage gain mainly depends on the opamp dc gain. By assuming A and C_p are 40dB and $2C_s$ respectively, the inter-stage gain has a step resolution around 0.14% when it approaches 4. The lowest opamp dc gain that can be calibrated back to 10b accuracy is 30dB in the 1st MDAC.

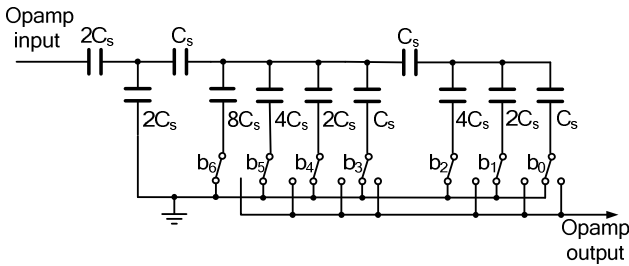


Fig. 5. Binary-weighted calibration capacitor.

C. Opamp

Fig. 6 presents a low-gain two-stage opamp used in the MDAC. To increase the opamp bandwidth, short channel devices are utilized for large transconductance and low parasitic capacitance. It results in about 20dB dc gain each stage. As mentioned above, the IOS technique is implemented to sample the opamp offset with unity-gain configuration during the self-calibrated process. An extra compensation capacitor is added for better phase margin by setting $CALb$ to be low.

The opamp in the 1st MDAC requires the largest bandwidth and consumes the highest power. The simulation results show that the opamp in the 1st MDAC has dc gain around 40dB. In the normal conversion, the closed-loop unity-gain bandwidth is 1.36GHz and the phase margin is 71.6 degrees with an output loading of 2pF. It consumes a DC current of 8mA. To save power, the bandwidth and current of the opamps in the following stages are scaled down accordingly.

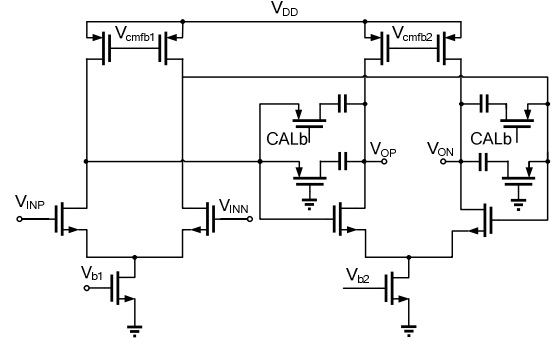


Fig. 6. Two-stage opamp.

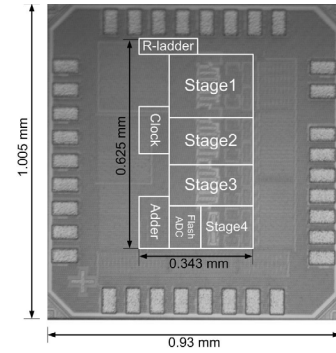


Fig. 7. Chip micrograph.

IV. EXPERIMENTAL RESULTS

The prototype ADC is fabricated in 90nm low-power CMOS technology, and its active area is 0.21mm^2 ($0.625\text{mm} \times 0.343\text{mm}$). The chip micrograph is illustrated in Fig. 7. The measured DNL and INL before calibration are $+1.7/-1.0$ LSB and $+15.6/-15.2$ LSB as shown in Fig. 8(a). After the self-calibrated process, the DNL and INL become $+0.7/-0.6$ LSB and $+0.8/-0.9$ LSB as in Fig. 8(b). The dynamic performance of varying input frequency at 320MS/s is presented in Fig. 9. The peak SNDR is 54.2dB. The

harmonic distortion starts to dominate the performance when input frequency is higher than 128MHz. At Nyquist rate input frequency, the SNDR is 51.2 dB, and the corresponding ERBW is 160MHz. Fig. 10 provides FFT plot for 320MS/s at input frequency of 128MHz. For the 1st MDAC, the calibration capacitance reaches its maximum value when the linearity performance of the ADC is the highest. It is limited by the under-estimated calibration range. Both the calibration parasitic capacitance and the opamp dc gain are worse than the expected values. The total power dissipation is 42mW, where 7.5mW, 1.5mW and 33mW are consumed by clock generator, digital and analog circuits respectively. The FOM, defined as $\text{Power}/(2^{\text{ENOB}} \times F_s)$, is 442fJ/conv.-step where ENOB is obtained at Nyquist rate. Finally, Table I summarizes and compares the measurement results with state-of-the-art single-channel high-speed 10b ADCs.

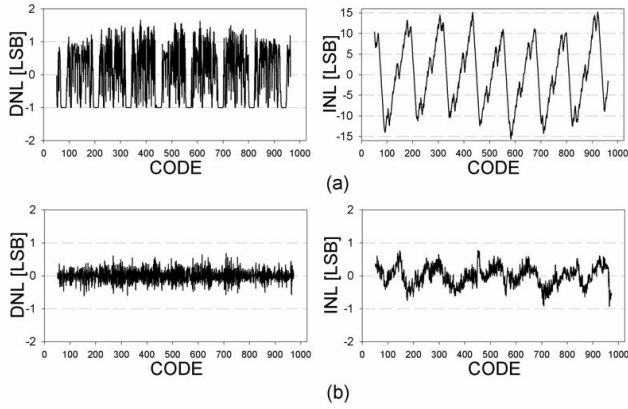


Fig. 8. DNL and INL performance (a) before calibration (b) after calibration.

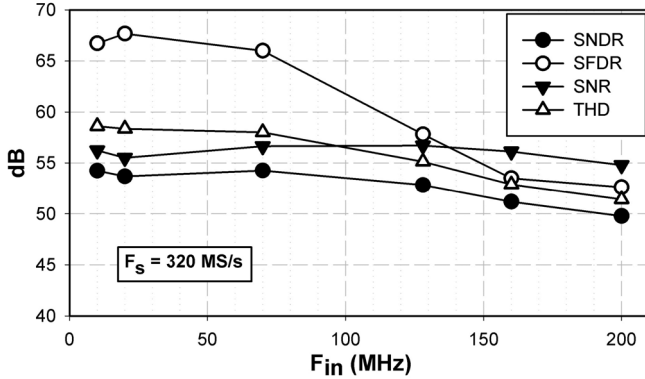


Fig. 9. Dynamic performance vs. input frequency at 320MS/s.

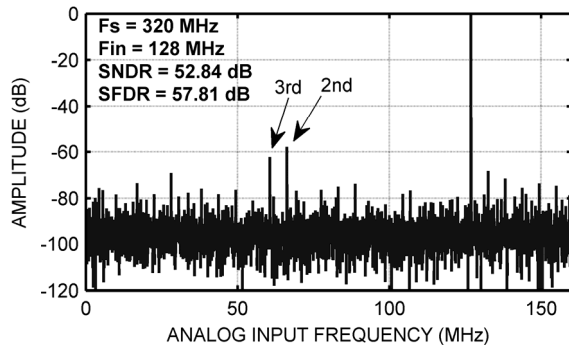


Fig. 10. FFT plot for 320MS/s at 128MHz input frequency.

	JSSC09 [1]	ISSCC07 [2]	This Work
Technology (nm)	90	130	90
Calibration Method	Foreground	Foreground /Background	Foreground
Sample Rate (MS/s)	500	205	320
Resolution (bit)	10	10	10
DNL/INL (LSB)	0.4/1.0	0.15/0.6	0.7/0.9
Peak SNDR (dB)	55.8	56	54.2
SNDR (dB) at $F_s/2$	53	56	51.2
SFDR (dB)	-	73.5	66.7
Power (mW)	55	92.5	42
FoM (fJ/c.-s.)	301	881	442
Active Area (mm ²)	0.49	0.52	0.21
Note	Calibration circuit is off-chip	Input buffer power is included	

Table I. Measured performance summary and comparison.

V. CONCLUSION

This paper proposes a self-calibrated technique by reducing feedback factor. The gain error caused by low gain opamp is corrected. Since the self-calibrated process does not need external precise reference sources and it can be done within a very short time, the proposed calibration can be processed easily. It achieves 8.7 ENOB at conversion rate of 320MS/s and only consumes 42mW that is able to fulfill the high bandwidth requirement for multimedia and wireless systems.

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